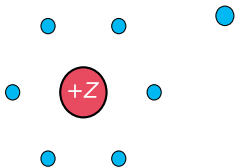


All-electron description



nucleus and core electrons
are explicit

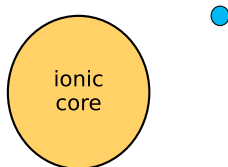
bare attraction $v_{\text{ext}} \sim -Z/r$

~~singular~~

replace



Pseudopotential description



only valence electrons
are explicit
 $Z_{\text{ion}} = Z - N_{\text{core}}$

effective operator $\hat{V}_{\text{GTH}}$

~~smooth core~~